Problem

Solve the following two equations by matrix inversion.

$2y_1 - y_2 = 4, \quad y_1 + 3y_2 = 9$

Step-by-step solution

Step 1 of 3

Consider the following set of equations,

$$2y_1 - y_2 = 4$$
$$y_1 + 3y_2 = 9$$

Arrange the equations in matrix form.

$$\underbrace{\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}}_{\mathbf{A}} \underbrace{\begin{bmatrix} y_1 \\ y_2 \end{bmatrix}}_{\mathbf{Y}} = \underbrace{\begin{bmatrix} 4 \\ 9 \end{bmatrix}}_{\mathbf{B}}$$

The matrix equation is,

$$\mathbf{AY} = \mathbf{B}$$

Here, $\mathbf{A} = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \quad \mathbf{B} = \begin{bmatrix} 4 \\ 9 \end{bmatrix} \quad \mathbf{Y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$

Step 2 of 3

From the matrix equation, $\mathbf{AY} = \mathbf{B}$, system of equations are solved by invert the matrix **A** and multiply it with **B**.

$$\mathbf{Y} = \mathbf{A}^{-1}\mathbf{B}$$

Calculate the determinant of matrix **A**, Δ.

$$\begin{aligned} \Delta &= \begin{vmatrix} 2 & -1 \\ 1 & 3 \end{vmatrix} \\ &= [2(3)] + [1(1)] \\ &= 6 + 1 \\ &= 7 \end{aligned}$$

Determine the inverse of matrix **A**.

$$\begin{aligned} \mathbf{A}^{-1} &= \frac{1}{|\mathbf{A}|} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \\ &= \frac{1}{7} \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \end{aligned}$$

Step 3 of 3

Calculate the system matrix **Y**.

$$
\begin{aligned}
\mathbf{Y} &= \mathbf{A}^{-1}\mathbf{B} \\
&= \frac{1}{7}\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}\begin{bmatrix} 4 \\ 9 \end{bmatrix} \\
&= \frac{1}{7}\begin{bmatrix} 12+9 \\ -4+18 \end{bmatrix} \\
&= \begin{bmatrix} 3 \\ 2 \end{bmatrix}
\end{aligned}
$$

Thus, the solution to the simultaneous equations is $\boxed{y_1 = 3,\ y_2 = 2}$.